INCH-POUND

MIL-PRF-1/7J
4 January 2008
SUPERSEDING
MIL-E-1/7H
17 February 1981

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5726

Inactive for new design
after 7 March 1997

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Requirements for acquiring the electron tube described herein shall consist of this specification sheet and MIL-PRF-1.

DESCRIPTION: Twin diode, miniature.

Outline --- 5-1 (EIA).

Base -- - E7-1.

Envelope --- T5-1/2.

Cathode --- Coated unipotential.

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	1k	2a	h	h	2k	sd	1a

ABSOLUTE RATINGS:

Parameter:	Ef	Epp/a	epx	Ehk	R1	C1	Io/a	ib/a	i surge/a	TE	Alt
Unit:	V	V ac	v	v	Ohms	µF	mA dc	ma	ma	°C	ft
Maximum:	6.9	---	360 (see note 1)	360	---	---	9.0	54	320	+165	Note 2
Minimum:	5.7	---	---	---	---	---	---	---	---	---	---
Test conditions:	6.3	165	---	---	11,000	8	---	---	---	---	---

GENERAL:

Qualification: Not required.
Reliable tube.

AMSC N/A FSC 5960

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TABLE I. Inspection tests.

MIL-STD-1311 method	Requirement or test	Conditions	Acceptance level (see note 11)	Symbol	Limits		Unit
					Min	Max	
	Conformance inspection, part 1						
1301	Heater current		0.4	If	275	325	mA
1336	Heater-cathode leakage	See note 3	0.4	Ihk	---	5	µA dc
1353	Operation of rectifiers	See notes 4 and 5	0.4	Io	16.0	20.0	mA dc
1231	Emission (1)	Eb = 10 V dc (see notes 3 and 5)	0.4	Is	40	---	mA dc
1201	Short and discontinuity detection		0.4	---	---	---	---
	Conformance inspection, part 2						
1211	Insulation of electrodes	E(a-all) = -300 V dc E(sd-all) = -300 V dc (see note 3)	2.5	R	200	---	MegΩ
1231	Emission (2)	See notes 3 and 6	2.5	ΔIs Ef	---	15	%
1256	Electrode current (anode)	Ebb = 0; Rp = 40,000 ohms (see note 3)	2.5	Ib	2	20	µA dc
---	Electrode current (anode) (difference between diodes)	Ebb = 0; Rp = 40,000 ohms (see note 3)	2.5	Ib	---	5	µA dc
---	Hum	Ef = 7.0 V ac (see note 7)	2.5	E	---	10	mV ac
1331	Direct-interelectrode capacitance	Shield No. 316 all measurements 1a to 2a 1a to h + 1k + shields 2a to h +2k + shields 1k to h +1a + shields 2k to h +2a + shields	6.5	 C C C C C	 --- 2.4 2.4 3.1 3.1	 0.026 4.0 4.0 4.7 4.7	 pF pF pF pF pF

See notes at end of table.

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TABLE I. Inspection tests - Continued.

MIL-STD-1311 method	Requirement or test	Conditions	Acceptance level (see note 11)	Symbol	Limits		Unit
					Min	Max	
	Conformance inspection, part 2 - Continued.						
1031	High-frequency vibration	No voltages	6.5	---	---	---	---
1041	Shock	675 G; Epp/a = 0	6.5	---	---	---	---
1031	Vibration-fatigue		6.5	---	---	---	---
---	Shock and vibration - fatigue test end points:						
1336	Heater-cathode leakage		---	Ihk	---	15	µA dc
1353	Operation of rectifiers		---	Io	14	---	mA dc
1121	Base strain		---	---	---	---	---
2126	Envelope strain		2.5	---	---	---	---
1105	Permanence of marking		---	---	---	---	---

MIL-STD-1311 method	Requirement or test	Conditions	Symbol	Limits		Unit
				Min	Max	
	Conformance inspection, part 3					
1506	Heater-cycling life	Ef = 7.5 V; Ehk = 135 V dc; Eb = 0	---	---	---	---
---	Heater-cycling life-test end point:					
1336	Heater-cathode leakage		Ihk	---	20	µA dc
1501	Intermittent life-test	TE = +165°C (min) (see notes 9 and 10)	---	---	---	---

See notes at end of table.

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TABLE I. Inspection tests - Continued.

MIL-STD-1311 method	Requirement or test	Conditions	Symbol	Limits		Unit
				MIN	MAX	
	Conformance inspection, part 3 - Continued					
---	Intermittent life-test end points (500 hours):	See note 11				
1301	Heater current		If	275	330	mA
1231	Emission (1)		Is	35	---	mA dc
1231	Emission (2)		ΔIs Ef	---	20	%
1336	Heater-cathode leakage		Ihk	---	10	μA dc
1211	Insulation of electrodes		R	100	---	MegΩ
---	Intermittent life-test end points (1,000 hours):	See note 11				
1231	Heater current		If	275	333	mA
1231	Emission (1)		Is	30	---	mA dc
	Emission (2)		ΔIs	---	25	%
1336			Ef			
1211	Heater-cathode leakage		Ihk	---	10	μA dc
---	Insulation of electrodes		R	75	---	MegΩ
			---	---	---	---

NOTES:

1. The maximum voltage appearing between any pair of pins shall be no greater than the peak inverse anode voltage rating.
2. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
3. Test each unit separately.
4. In a full-wave circuit, adjust Zp/a so that a bogey tube gives Io = 18 mA dc. A bogey tube has a voltage drop of Etd = 10 V dc at Is = 60 mA dc per anode. Ehk = Eo + 117 V ac.
5. This test shall be performed at the conclusion of the holding period.
6. Test at Eb = 7.0 V ac with readings at Ef = 6.3 V and Ef = 5.7 V.

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7. The test circuit to be used is shown in figure 1.

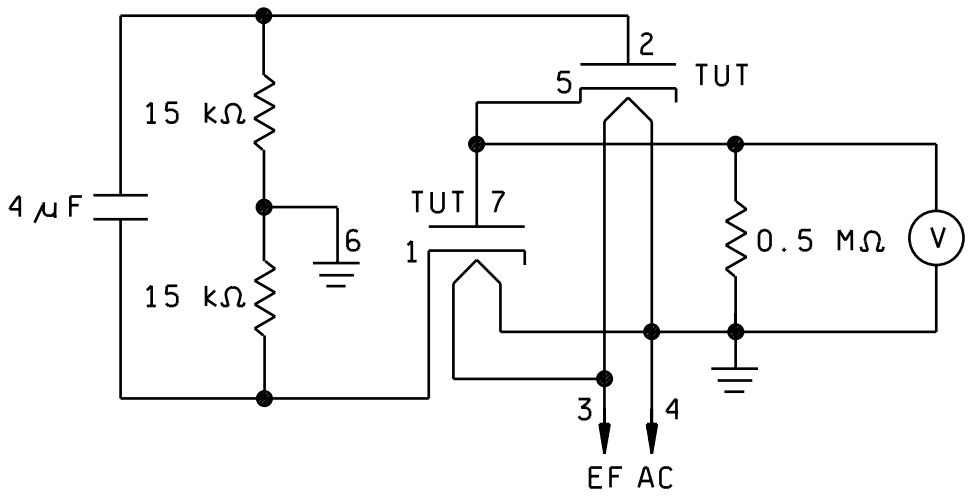


FIGURE 1. Test circuit.

8. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes.

9. In a full-wave life-test circuit, the values specified for R1 and C1 may be considered as approximate and shall be adjusted initially to give not less than Io = 18 mA dc and ib = 50 ma per anode with a bogey tube. Ehk = Eo + 117 V ac.

10. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (MIL-STD-1311, method 1226), will be satisfied if a tube having Etd = 10 V dc (±5 percent) at Is = 60 mA dc per anode, under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

11. This specification sheet uses accept on zero defect sampling in accordance with MIL-PRF-1, table III.

12. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Referenced documents: In addition to MIL-PRF-1, this specification sheet references MIL-STD-1311.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5960-2008-003)

Review activities:
Army - AR
Navy - AS, CG, MC, OS
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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